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Targeted enzyme discovery using metal-coordination mining

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The recent revolution in genome sequencing and protein structure prediction has opened new frontiers in understanding, predicting and designing enzyme function^{1,2}. Central to these efforts is the discovery and functional annotation of novel enzymes, which is essential for elucidating the connection between genotype and phenotype and for developing biocatalysts for industrial applications. However, accurately predicting enzymatic function remains a major challenge, and the discovery of new enzymes often relies on serendipity. Here we present a metal-coordination-guided strategy that uses atomic-level mechanistic principles to mine protein structure databases for the targeted discovery of metalloenzymes. We apply this framework to the AlphaFold2 Protein Structure Database to identify new members of the Fe^{II}/α-ketoglutarate-dependent halogenase family, which selectively functionalize unactivated C(sp³)–H-bonds, a crucial transformation in the production of pharmaceuticals and other high-value compounds^{3,4}. These radical halogenases constitute a low-abundance class within the large and diverse cupin superfamily⁵. Owing to low sequence conservation, they have been especially challenging to find against the complex background of related family members, such as hydroxylases, desaturases and epimerases. Our metal-coordination mining methodology reveals several previously unrecognized radical halogenase families spanning diverse phylogenetic space, at minimal computational cost. Our predictions are validated by the experimental characterization of two new radical halogenases, AspX and BtnX. Notably, BtnX shows a substrate promiscuity that is unprecedented in radical halogenases, opening the way for a broad range of biocatalytic applications.

Main

Enzymes have a central role in living systems, catalysing on the order of 10⁵ chemical transformations that are needed to sustain cellular metabolism. This catalytic plasticity has been driven by the divergent evolution of large superfamilies containing members that are phylogenetically related yet have diverse functions^{6,7,8}. Accurate functional annotation of individual enzymes is crucial for reaching the level of precision needed to enable both

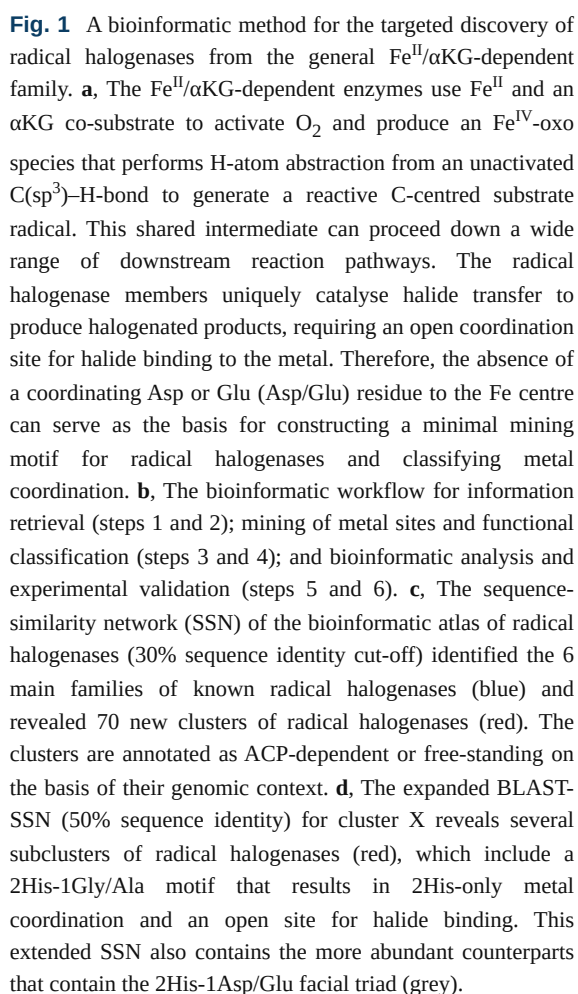
the discovery of new enzymes and the prediction of biological function at an organismal level. Sequence-similarity-based methods take advantage of the vast quantity of DNA sequence available^{9,10,11}; however, the evolution of gene sequence is not dictated just by enzymatic function. Instead, it is confounded by many factors, including positive and negative selection, neutral drift, mutation rate, redundancy, dispensability, network organization, metabolic burden and expression level¹². As a result, the primary sequence of an enzyme reflects a complex history of convergent and divergent

evolution as well as interspecies transfer and phylogenetic relationships. By contrast, enzyme structure–function relationships are expected to be much more highly conserved because many elements, such as catalytic architecture, substrate-binding pockets, thermodynamic stability, protein dynamics and protein fold, need to be maintained for functionality^{7,13,14}.

Advances in protein structure prediction have enabled the development of large databases, such as the AlphaFold2 (AF2) Protein Structure Database¹ and the ESM Metagenomic Atlas¹⁵, that offer new opportunities for structure-based discovery. Indeed, domain-based discovery approaches can be applied successfully to specialized structural folds that have been co-opted for specific functions^{16,17,18}. However, it is more common to find that such stable folds have been recruited repeatedly throughout evolution and can thus support an extremely broad range of functions, making it difficult to make precise predictions¹⁹. In this regard, distinct enzymatic functions within a structural fold should give rise to conserved differences in the chemical reaction mechanism that can be correlated with atomic-level structural divergence in the active site^{6,7,8,14}. This conceptual framework suggests that key mechanistic information could be encoded in atomistic representations of the active-site chemistry and guide the large-scale mining of protein structure databases for targeted discovery of enzymes with desired functions.

Towards this goal, we have developed a pipeline for the targeted discovery of metalloenzymes. It is estimated that 25–50% of proteins require a metal ion for their function²⁰, using the tunable redox, acid–base and structural properties of metal ions to expand the reaction space beyond what is possible with the standard amino acid functional groups. There has been great interest in the computational identification of these sites using approaches such as co-evolution or machine learning^{21,22,23,24}, but these approaches are currently limited in their ability to predict specific enzymatic functions because of the lack of validated functional annotations, which are needed for accurate training. However, biological metal sites are governed by specific geometric constraints related directly to

their catalytic mechanisms, such as those defined by distinct metal-coordination bonds with protein-derived ligands, second-sphere interactions and hydrogen-bond networks. Thus, these types of mechanistically relevant interactions establish three-dimensional (3D) motifs that can be used for the targeted structural mining of specific metal-binding sites as well as for their accurate functional classification, even within the context of highly diverse protein superfamilies. This metal-coordination structural mining approach was successfully implemented and experimentally validated towards the discovery of new Fe^{II}/α-ketoglutarate (αKG)-dependent radical halogenases²⁵, which are low-abundance members of the cupin superfamily that catalyse a synthetically useful C–H functionalization reaction, replacing inert C(sp³)–H-bonds with native (X = Cl) and non-native (X = Br, N₃) anions^{26,27} (Fig. 1a).



Certain protein architectures have been found to be highly recurrent, arising multiple times in evolution. These superfolds are used by many distinct families with diverse functions. The cupin superfold is found in all domains of life and exhibits an exceptionally wide range of biochemical functions—on par with the versatility of the triose phosphate isomerase (TIM)

The Fe^{II}/αKG-dependent enzymes use the cupin superfold and participate in a broad array of metabolic transformations, from the modification of proteins to key steps in the biosynthesis of antibiotics and other natural products. These enzymes use a mononuclear non-haem Fe^{II} cofactor and the αKG co-substrate to bind to and activate O₂ and generate a reactive Fe^{IV}-oxo intermediate³³ (Fig. 1a). However, this shared intermediate can catalyse a wide range of reactions, such as hydroxylation, desaturation, cyclization and halogenation³³ (Extended Data Fig. 1). The prototypical members of this family bind to Fe^{II} using a cupin-derived 2His-1Asp/Glu ‘facial triad’ motif and the αKG co-substrate to form the primary coordination sphere of the metal³³. The radical halogenase members are distributed at low frequency within various families of Fe^{II}/αKG-dependent enzymes, but are mechanistically distinct with regard to their ability to transfer an Fe-bound anion to the substrate⁵. This step necessitates the presence of an open coordination site for halide coordination to Fe^{II}, which is observed as the conversion of the 2His-1Asp/Glu motif to the non-

canonical 2His-1Gly/Ala motif in all previously reported members (Fig. 1a and Extended Data Fig. 2). Thus, we reasoned that this structure–function relationship, in which metal coordination dictates reaction outcomes, could serve as a bioinformatic marker for the identification of radical halogenases.

Sequence-based methods typically rely on pairwise comparisons that scale as N^2 (where N is the number of sequences) and thus lack both the sensitivity and the scalability to search for a single amino acid change within a large superfamily where $N > 10^5$. To address these limitations, we turned to mechanism-guided structural bioinformatics, with the rationale of using the atomic-level 3D features of the metal coordination in radical halogenases: (i) the 2His coordination motif, which is shared across all members in the superfamily, as a simple structural representation to identify the Fe-binding site; and (ii) the absence of additional coordinating ligands, such as the canonical Asp or Glu residues, for the functional classification of a metal-binding structure as a radical halogenase (Fig. 1a). This approach addresses both the challenge of sensitivity, by using geometric clustering rules in 3D space to improve on the lower detection signal from untargeted alignment scores in one-dimensional (1D) space, and that of scalability, by using a structural motif search that scales as N^1 .

A structural database of the $\text{Fe}^{\text{II}}/\alpha\text{KG}$ -dependent enzyme family was compiled by first retrieving protein sequences annotated to contain the cupin domain ($n = 1.8$ million) from the approximately 220 million sequences in the InterPro database (Fig. 1b and Supplementary Table 2). After an initial filtering step to remove fragments, duplicates and non-enzymes, the corresponding structural models from the AF2 Protein Structure Database were retrieved ($n = 530,814$) and mined for the presence of a 2His binding site using the unique distance and orientation relationship that would allow two intra-strand His residues to act as a metal-binding site³⁴. We found that three constraints were sufficient to describe this site, with two H-bonds between the backbone amide and carbonyl groups from His_A and His_B defining their location on adjacent β -strands, and the side-chain N τ –N τ distance defining

their preorganization for metal coordination (Fig. 1a). Most of these predicted structures contained a 2His metal-binding site with an additional Asp/Glu residue in coordination proximity to the metal site ($n = 456,585$); a small minority lacked a nearby coordinating residue and instead contained an Ala/Gly residue ($n = 946$), suggesting radical halogenase activity (Fig. 1b).

The sequence-similarity network (SSN) of these predicted radical halogenases includes all previously known and experimentally verified radical halogenases^{5,25,35,36} (Fig. 1c), which serves as an initial positive validation of our methodology. This SSN captures radical halogenase clusters (greater than 30% sequence identity) with much higher coverage than what can be achieved using sequence-based methods alone, as illustrated by dechloroacetamine halogenase (DAH), a rare case of a eukaryotic halogenase³⁷. The previously reported sequence homology searches identified only a single DAH homologue³⁷, consistent with our own BLAST search coupled with multiple sequence alignment, which identified only three homologues (Supplementary Fig. 1). By contrast, our metal-coordination mining approach defined a DAH cluster comprising 40 putative radical halogenases from both eukaryotic (fungal and plant) and bacterial origins (Fig. 1c). More broadly, the SSN for the complete bioinformatic atlas of radical halogenases contains 70 clusters that seem to be completely unexplored, with no previous radical halogenase annotation, and which span a vast taxonomic and phylogenetic protein sequence space, including several new families of eukaryotic halogenases (Fig. 1c and Extended Data Fig. 3). This expanded and refined bioinformatic atlas of radical halogenases (Supplementary Data 1) provides a valuable resource for the wider scientific community that can enable novel chemical transformations to be investigated in unique biological contexts and can accelerate the discovery of new biocatalysts with existing experimental high-throughput pipelines³⁸.

Discovery of new radical halogenases

Motivated to further validate our metalloenzyme discovery platform and apply it towards discovering $\text{Fe}^{\text{II}}/\alpha\text{KG}$ -dependent halogenases, we selected a new family from our SSN (cluster X) for experimental validation (Fig. 1c). Cluster X was intriguing because of the diverse genomic context of its putative radical halogenase members; some were co-clustered with acyl-carrier protein (ACP)-related genes, indicating that they could be ACP-dependent radical halogenases, whereas others lacked these neighbouring genes, suggesting that they use free-standing substrates (Supplementary Fig. 2). Sequence enrichment of cluster X with all phylogenetically related sequences through BLAST (E-value = 5), followed by generation of a more granular SSN (greater than 50% sequence identity)³⁹, reveals many new subfamilies of radical halogenases (Fig. 1d). This expanded SSN shows a mixed distribution of halogenases and their non-halogenase counterparts, suggesting complex evolutionary trajectories that mirror the broader empirical trend across the $\text{Fe}^{\text{II}}/\alpha\text{KG}$ -dependent family. Although this complex evolution poses a considerable challenge in sequence-based enzyme discovery, given the differing locations of the three residues of the 2His-1Ala/Gly motif in the 1D sequence space, these issues are rapidly resolved in the 3D search space¹⁹.

An analysis of the genome neighbourhoods of radical halogenase candidates in cluster X revealed subfamilies in which the genes of their ACP-independent radical halogenase members are adjacent to those encoding amino acid transporters and amino-acid-modifying enzymes (for example, ATP-grasp ligases). On the basis of this observation, we hypothesized that some of these radical halogenases act on free amino acids. A putative radical halogenase from the marine pathogen *Vibrio campbelli*, AspX, was selected from one of these subclusters for biochemical validation (Fig. 2a and Supplementary Table 3). The in vitro reaction of AspX with a mixture of the 20 canonical L-amino acids analysed by liquid chromatography–quadrupole time-of-flight mass spectrometry (LC–QTOF) shows that AspX selectively chlorinates L-aspartic acid but does not act on any of the other amino acids (Fig. 2b and Supplementary Fig. 3). Nuclear magnetic resonance (NMR)

characterization of the isolated AspX product (Supplementary Fig. 4) and steady-state kinetics show that AspX converts L-aspartic acid to 3S-chloro-L-aspartic acid (catalytic constant (k_{cat}) = $33.3 \pm 0.5 \text{ min}^{-1}$, Michaelis constant (K_{m}) = $0.64 \pm 0.02 \text{ mM}$, turnover number = 780 ± 107 ; Extended Data Fig. 4). AspX can also transfer alternative anions (Br^- and N_3^-) to produce bromo- and azido-L-aspartic acid derivatives (Extended Data Fig. 5a). The discovery of AspX extends the previously reported scope of free amino acid halogenases acting on positively charged (L-lysine and L-ornithine) and non-polar (L-leucine, L-isoleucine and L-norleucine) amino acids⁵ to the first case of a negatively charged (L-aspartic acid) amino acid.

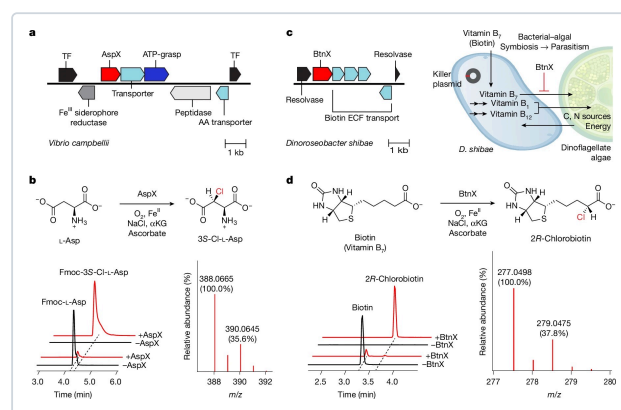


Fig. 2 Discovery of a family of α -carboxylic acid radical halogenases. **a**, AspX is part of a bacterial biosynthetic gene cluster that contains genes related to amino acid modification (ATP-grasp) and amino acid (AA) transport. TF, transcription factor. **b**, In vitro conversion of L-aspartate (L-Asp) to 3S-chloro-L-aspartate (3S-Cl-L-Asp) by AspX. AspX (25 μM) was incubated aerobically with L-Asp (500 μM) in 50 mM HEPES (pH 7.5) containing 1 mM Fe^{II} , 5 mM αKG , 5 mM ascorbate and 100 mM NaCl for 2 h. The product was derivatized with 9-fluorenylmethyloxycarbonyl (Fmoc-Cl) before analysing by LC-QTOF (Fmoc-L-Asp, m/z = 354.0983; Fmoc-Cl-L-Asp, m/z = 388.0593). **c**, BtnX is co-localized with an energy-coupling factor (ECF) transporter complex for biotin uptake that is located on a ‘killer plasmid’ and upregulated during bacterial–algal parasitism. **d**, In vitro conversion of biotin to 2R-chlorobiotin by BtnX. BtnX (25 μM) was incubated aerobically with biotin (500 μM) in 50 mM HEPES (pH 7.5) containing 1 mM Fe^{II} , 5 mM αKG , 5 mM ascorbate and 100 mM NaCl for 2 h before analysing by LC-QTOF (biotin, m/z = 243.0809; 2R-chlorobiotin, m/z = 277.0419). Chromatograms are individual traces for the calculated m/z values that are representative of technical replicates ($n = 3$).

Another subfamily of cluster X is located at the junction of ACP-dependent and free-standing halogenases (Fig. 1d). Although its members share some sequence similarity with the known ACP-dependent halogenase CurA (around 25–30% sequence identity)⁴⁰, many of their associated biosynthetic gene clusters lack genes associated with ACP-dependent halogenases, raising the possibility that they accept alternative—and possibly large—substrates (Supplementary Fig. 2). A sequence homology search against the published literature (PaperBLAST⁴¹) for members of this subfamily revealed a possible biological role for one of its putative radical halogenase members, BtnX, which is found on the ‘killer plasmid’ of the marine bacterium *Dinoroseobacter shibae*⁴² (Fig. 2c). *D. shibae* exists in a symbiotic relationship with the dinoflagellate algae *Prorocentrum minimum*, in which the bacterium provides vitamins and other micronutrients to the algal host in return for energy, carbon and nitrogen. These bacterial–algal symbionts grow to form red toxic algal blooms that can expand to large geobiological scales and exert major negative effects on entire ecosystems⁴². The presence of the killer plasmid has been directly implicated in turning on a pathogenic phase in *D. shibae*, in which it kill its algal host through micronutrient starvation and adopts a metabolic strategy of feeding off algal senescence⁴³. Although the biochemical and molecular origins of this biological phenomenon remain unclear, gene knockout experiments indicate that Dshi_3684, the gene that encodes BtnX, is positively associated with bacterial pathogenicity⁴². Notably, Dshi_3684 is co-localized with genes that encode a putative energy-coupling factor (ECF) biotin transporter complex (Fig. 2c). The combined evidence gave rise to the hypothesis that BtnX is a radical halogenase acting on biotin, also known as vitamin B7. The in vitro reaction of BtnX with biotin (Fig. 2d and Supplementary Table 3) and product characterization by NMR (Supplementary Fig. 5) show that BtnX converts biotin to 2*R*-chlorobiotin. Steady-state kinetics reveal that BtnX has a strong binding affinity for biotin ($k_{\text{cat}} = 0.96 \pm 0.02 \text{ min}^{-1}$, $K_{\text{m}} < 2 \text{ }\mu\text{M}$, turnover number = 22 ± 6 ; Extended Data Fig. 4b), consistent with the low levels of biotin in

marine environments⁴⁴ (0.7 nM to less than 1.2 pM). The stereoselectivity of the BtnX chlorination reaction was confirmed by X-ray crystallographic investigations on the aerobic product-bound structure of BtnX (Protein Data Bank (PDB) ID: 9Q04; Supplementary Table 4 and Extended Data Fig. 6). Similarly to AspX and other known radical halogenases^{5,26}, BtnX can transfer alternative anions (Br^- and N_3^-) to produce bromo- and azido-biotin derivatives (Extended Data Fig. 5b). Collectively, on the basis of its strong binding affinity as well as the genomic and biological context, we propose that biotin is the native substrate of BtnX. The discovery and characterization of BtnX increases the known substrate range of radical halogenases beyond previously identified classes to include the functionalization of vitamins.

A versatile radical halogenase

Whereas AspX and all other previously known radical halogenases exhibit relatively narrow substrate scopes that are often limited to their respective native substrates^{5,36,45}, BtnX seems to be a highly promiscuous radical halogenase, catalysing the chlorination of a wide range of non-native substrates (Fig. 3). Structure–activity studies reveal that the presence of a propionate head group is the key requirement for substrate recognition and halogenation by BtnX, whereas chemical and structural variation on the substrate ‘tail’ is highly tolerated (Supplementary Fig. 6). Therefore, BtnX can enable access to a wide range of regio- and stereoselectively functionalized molecules, from free and functionalized acids (including bile acids and fluorescent dyes) to amino acids, peptoids and peptides. These transformations are relevant to both the early-stage functionalization of simple molecules to produce α -halo acid precursors and the late-stage derivatization of pharmaceuticals, as demonstrated by the enzymatic halogenation of RLA8, a drug candidate for the treatment of nonalcoholic steatohepatitis and fibrosis⁴⁶.

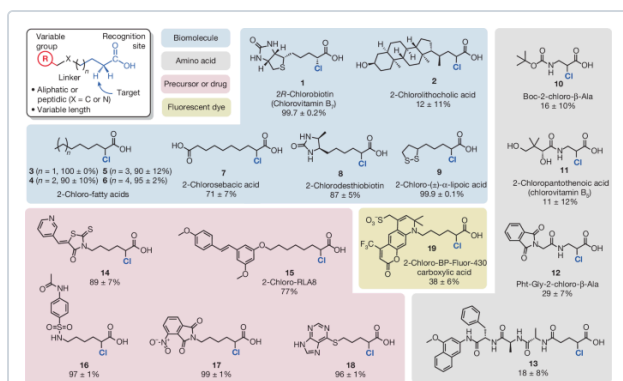


Fig. 3 BtnX is a promiscuous radical halogenase for the production of α -halo acids. The substrate scope of BtnX ranges from free fatty acids to various carboxylic- and amino-acid-containing biological and synthetic molecules. The substrate consumption yields (%) were determined by high-resolution LC-QTOF and represent mean $\pm$ s.d. ($n = 3$ technical replicates; Supplementary Fig. 6e). All reactions were performed at room temperature for 3 h with 5 mol% BtnX added.

To identify the molecular basis for this promiscuity, we solved the anaerobic crystal structure of BtnX bound to biotin in the early stages of its catalytic reaction mechanism. The crystal structure of BtnX (PDB ID: 9PV1; Supplementary Table 5) was solved at 1.20 Å in the presence of its native ligands (biotin, α KG and chloride), and Fe^{II} was added under anaerobic conditions. The asymmetric unit contains a BtnX dimer in which the active site of chain A contains α KG and biotin and the active site of chain B contains Fe^{II} , α KG, Cl and biotin (Fig. 4a, Extended Data Fig. 7 and Supplementary Fig. 7). This structure reveals two key features of the ability of BtnX to accept such varied substrates. First, the carboxylate head group of biotin seems to form specific interactions within the substrate cavity, through multiple H-bonds with active-site residues (Ser120, Asn119 and Arg248) that position the pro-2R H atom in close proximity to the metal site for stereoselective C–H functionalization (Fig. 4a). Indeed, perturbations of the H-bonding network around Ser120 by site-directed mutagenesis resulted in a decrease in both enzymatic reactivity and chemoselectivity (Extended Data Fig. 8). Second, the rest of the biotin substrate is located in an extended solvent-exposed protein channel and forms mostly non-specific hydrophobic interactions (Fig. 4b and Supplementary Fig. 8e). Together, the combination of

specific H-bonding interactions surrounding the target C–H site with a solvent-exposed channel forms the molecular basis of the substrate promiscuity of BtnX.

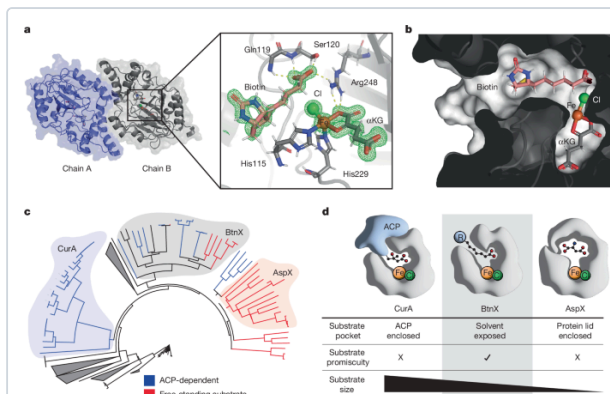


Fig. 4 Structural basis of the substrate promiscuity of BtnX.

a, BtnX crystallizes as a homodimer (1.20 Å, PDB ID: 9PV1). Chain B has a 2His-1Gly site that coordinates Fe^{II} , and also contains the bound Cl, α KG and biotin ($F_0 - F_c$ maps contoured at 2σ), whereas chain A contains only α KG and biotin (Extended Data Fig. 7). The holo active site in chain B also reveals key H-bonding interactions (yellow dashed lines) between the carboxylate group of the bound biotin substrate and active-site residues (Gln119, Ser120 and Arg248) and the α KG co-substrate. **b**, The biotin substrate binds in a mostly hydrophobic and solvent-exposed channel, which enables its broad substrate promiscuity. **c**, BtnX exists at a phylogenetic branch point between radical halogenases acting on ACP-bound substrates (CurA; blue) and those acting on free-standing substrates (AspX; red). **d**, Comparison of protein architectures of radical halogenase families and their structure–function relationships with respect to substrate promiscuity.

This protein architecture of BtnX is distinct from that of other radical halogenases, which typically contain buried active-site pockets and make multiple specific interactions across the entire substrate^{5,47} (Supplementary Fig. 8). The BtnX subcluster exists at a distinct phylogenetic branch point between ACP-dependent radical halogenases (CurA)⁴⁰ and radical halogenases that act on free-standing small molecules (AspX; Fig. 4c). Structural comparison reveals that, unlike CurA and AspX, the unique solvent-exposed channel of BtnX enables its promiscuous reactivity (Fig. 4d and Supplementary Fig. 8). By contrast, the substrate scope of the more typical members of the radical halogenase family has been more difficult to alter, given the tight geometric constraints enforced by the halogenation mechanism⁴⁸. As such, BtnX could

represent an unusually evolvable radical halogenase that is amenable to laboratory evolution for engineered applications, as well as serving as a template for the discovery of other promiscuous members across the broader Fe^{II}/αKG-dependent non-haem iron enzyme family.

The unique promiscuity of BtnX makes it a promising tool for applications in chemoenzymatic synthesis and biocatalysis⁴. In addition to the expanded substrate scope for chlorination, the chemical space accessed by BtnX extends even further when considering that BtnX can also transfer alternative anions (Br and N₃); brominated products are valuable precursors for further simple downstream modifications, such as amination reactions to produce a wide range of non-canonical amino acids. In addition to the biocatalytic utility of the wild-type BtnX, the introduction of G117D or G117E point mutations converts it into a hydroxylase for the production of α-hydroxy acids (Extended Data Fig. 9).

Conclusion

Although enzymes are universally recognized for their high selectivity as catalysts, their evolution is thought to be driven by widespread promiscuity with respect to substrate selectivity and reaction outcomes. This functional divergence is often controlled by a small number of specific residues near the catalytic centre, and cannot be captured by clustering methods based on total sequence similarity. As such, targeted enzyme discovery remains a long-standing challenge, despite its essential role in uncovering new biosynthetic potential in nature as well as in developing tools for biocatalysis and biotechnology. The emergence of accurate protein structure prediction algorithms, such as AF2, has generated large protein structure databases that now allow the use of specific 3D information to search and annotate enzymatic function^{16,17}. Here we show how fundamental mechanistic insights into enzyme function can be distilled into precise atomic-level constraints that describe functionally conserved physical features for mining the AF2 Protein Structure Database.

Despite the absence of metals and other cofactors in the AF2 structure database and in other related databases, the approach presented here uses the intrinsic preorganization of enzyme active sites for metal binding to mine predicted structures of metalloproteins in their fully apo forms. More broadly, metal-coordination-based mining provides a powerful route to uncover novel or rare enzymatic functions that have evolved through complex phylogenetic trajectories within large and diverse enzyme superfamilies. This capability enables the systematic discovery of uncommon catalytic activities that are difficult to access through conventional experimental approaches alone. In the case of Fe^{II}/αKG-dependent radical halogenases, 6 enzyme families have been discovered over the past 2 decades through natural products research, whereas our active-site mapping approach rapidly identified 70 previously unrecognized families of putative halogenases. Enzymatic halogenation of unactivated C(sp³)-H sites through radical chemistry is relatively rare in biology, where in addition to the Fe^{II}/αKG-dependent halogenases only two other classes of enzyme are known to catalyse such transformations: the binuclear iron⁴⁹ and the binuclear copper⁵⁰ halogenases (Extended Data Fig. 2).

Beyond targeted enzyme discovery, our approach makes it possible to uncover new metal-binding motifs and active sites using a fundamental chemical understanding of metal-protein interactions. To demonstrate that potential, we have extended our targeted discovery methodology to examine the diversity and distribution of 2His-X_n metal sites within the cupin superfold, which allows us to uncover both known and new types of putative biological metal site, with a wide range of coordination environments (Extended Data Fig. 10 and Supplementary Table 6). Notably, this methodology is scalable, owing to its algorithmic efficiency that does not require pairwise comparisons, and is thus ideal for managing the current explosion of sequence and structural information. Beyond enzymes, this approach can be extended to any protein with a distinct set of functionally important atomic interactions, such as those involved in recognition or signal transduction, bringing us one step

closer to understanding sequence–structure–function relationships at large scale.

Extended data figure legends

Extended Data Fig. 1 *Consensus catalytic mechanism and reactivity scope of Fe^{II}/αKG-dependent enzymes.* The consensus mechanism involves O₂ binding and activation to form a reactive high-spin (*S* = 2) Fe^{IV}-oxo intermediate that enables a diverse range of chemical transformations. Most enzymes in this family use the 2His-1Asp/Glu facial triad motif (blue box). However, in radical halogenases, the Asp/Glu ligand is replaced by a non-coordinating residue (Ala/Gly) that enables anion coordination (*X* = Cl), a key requirement for halogenation (red box).

Extended Data Fig. 2 *Representative members of Fe^{II}/αKG-dependent and other types of radical halogenases.* Fe/αKG-dependent radical halogenases from the previously characterized families act on substrates that are either bound to acyl carrier proteins as part of PKS, NRPS, or mixed PKS/NRPS assemblies or in their free-standing forms. The binuclear Fe halogenases (CylC) and binuclear Cu halogenases (ApnU) are also proposed to use radical reaction mechanisms.

Extended Data Fig. 3 *SSN of the bioinformatic atlas of radical halogenases.* The SSN for radical halogenases (Fig. 1c) is shown with the complete cluster numbering (1–68) and with taxonomic annotations (superkingdom and phylum) for each node.

Extended Data Fig. 4 *Steady-state kinetic characterization of AspX and BtnX.* **a**, The NADH-coupled assay used for measuring the steady-state kinetics by monitoring succinate formation (CoA, coenzyme A; SCS, succinyl-CoA synthetase; PEP, phosphoenolpyruvate; PK, pyruvate kinase; LDH, lactate dehydrogenase). **b**, Michaelis–Menten analysis for the reaction of AspX with L-aspartic acid ([AspX]₀ = 5.0 μM, pH = 7.5, 25 °C). **c**, The *V*_{max} dependence on [BtnX]₀ for the reaction of BtnX with biotin allows for the estimation of its *k*_{cat} parameter (<2 μM). **d**, Michaelis–Menten analysis for the reaction of BtnX with octanoic acid. The greater-than-tenfold increase in binding affinity of BtnX for biotin over octanoic acid implicates the role of the ureido ring of biotin in substrate–protein interactions.

Extended Data Fig. 5 *C–H functionalization with alternative anions by AspX and BtnX.* **a**, In addition to chlorination, AspX also catalyses the bromination and azidation of L-aspartic acid when chloride is replaced by bromide (100 mM) or azide (10 mM). **b**, BtnX likewise catalyses the bromination and azidation of biotin. Reactions were analysed by LC-QTOF in negative mode using a C18 column and are representative of *n* = 3 technical replicates.

Extended Data Fig. 6 *The aerobic BtnX structure contains the bound product formed in the single-turnover in-crystallo reaction.* **a**, The aerobic BtnX structure (1.96 Å, PDB ID: 9PV1) forms a dimer equivalent to the anaerobic structure. **b**, Two neighbouring loops near the αKG-binding site are disordered in chain B but not chain A. **c,d**, The active sites reveal crystallographic snapshots of product release intermediates. **e**, Proposed mechanism for chlorination of biotin and product release.

Extended Data Fig. 7 *Extended analysis of the anaerobic crystal structure of BtnX bound to Fe^{II}, αKG, chloride and biotin.* Includes the anomalous diffraction map consistent with Fe, chloride and biotin sulfur; single-wavelength anomalous scattering confirming the bound metal as Fe; a low-occupancy water ligand; and structural comparison of the chain A versus chain B active sites.

Extended Data Fig. 8 *Site-directed mutagenesis of BtnX to examine the contributions of active-site H-bonds in reactivity and chemoselectivity.* The biotin/S120/Y64/R248 H-bond network is crucial for both reactivity and chemoselectivity (halogenation versus hydroxylation). Statistical significance was assessed using two-sided unpaired Welch's *t*-tests with Benjamini–Hochberg correction; adjusted *P* values are reported.

Extended Data Fig. 9 *Engineering BtnX into an α-carboxylic acid radical hydroxylase.* Introduction of the G117D or G117E mutation fully suppresses halogenation while increasing hydroxylation by 40 ± 9-fold and 18 ± 6-fold, respectively.

Extended Data Fig. 10 *Diversity and distribution of 2His-*X*_n sites.* The SSN from representative sampling (*n*_{SSN} = 3,000) across the total 2His metal-binding sites in the cupin superfold (*n*_{total} = 538,160) shows a wide diversity of 2His-*X*_n metal sites with different numbers (*n* = 0, 1, 2, 3, 4) and identities (*X* = Asp, Glu, His, Asn, Gln, Cys, Met, Tyr, or none) of ligands.

Data and code availability

Data availability. Accession codes and sequences for AspX (UniProt ID A0AAC9SM19), BtnX (UniProt ID A8LT50) and all other proteins analysed in this study are provided in Supplementary Table 3. The structures of BtnX under anaerobic conditions in the presence of biotin, αKG, chloride and Fe^{II} (PDB ID 9PV1) and after aerobic turnover (PDB ID 9Q04) have been deposited in the Protein Data Bank. Pfam IDs used in the bioinformatic analysis are listed in Supplementary Table 2. All protein sequence and structure data were obtained from the open-access InterPro database and AlphaFold Protein Structure Database. Source data are provided with this paper.

Code availability. The complete Python (v.3.9.6) code used for the bioinformatic analyses is available as a Jupyter notebook via Zenodo at <https://doi.org/10.5281/zenodo.19737459>. The core functionality for identifying and annotating 2His and 2His-1Asp/Glu metal-binding motifs is also provided as a stand-alone Python package, freely available at <https://github.com/yannikipouros/hal-discovery>. The repository includes installation instructions, usage documentation and a demo dataset.

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Author information

Contributions. I.K. conceptualized the study, designed experiments, collected and analysed data and wrote the manuscript. M.C.Y.C. conceptualized the study, designed experiments, analysed data, supervised the project and wrote the manuscript. **Corresponding authors:** correspondence to Ioannis Kipouros or Michelle C. Y. Chang.

Competing interests. A patent application related to biocatalytic applications of BtnX has been filed by Princeton University, on which I.K. and M.C.Y.C. are inventors.

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